Task – 01: Implement Caesar Cipher

1. Introduction
The Caesar Cipher is one of the oldest and simplest encryption techniques. It is a type of substitution cipher in which each letter in the plaintext is shifted by a fixed number of positions in the alphabet. This technique was named after Julius Caesar, who reportedly used it for his private correspondence.

2. Objective
To create a Python program that can **encrypt and decrypt text** using the Caesar Cipher algorithm. The program should:
- Allow users to input a message.
- Allow users to input a shift value (key).
- Perform encryption or decryption based on user choice.

3. Algorithm
Step 1: Define an alphabet set (A–Z and symbols).
Step 2: Take input from the user:
  - Mode (Encrypt/Decrypt)
  - Key (Shift value)
  - Message
Step 3: For each character in the message:
  - Find its position in the alphabet.
  - If Encrypt, shift right by key positions.
  - If Decrypt, shift left by key positions.
  - Handle wrap-around using modulo arithmetic.
Step 4: Return the encrypted or decrypted message.

4. Python Implementation (Code)

```
ALPHABET = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ !@#$%^&*'

def caesar_cipher(message, key, mode):
    result = ""
    for ch in message.upper():
        if ch in ALPHABET:
            idx = ALPHABET.index(ch)
            if mode == "encrypt":
                idx = (idx + key) % len(ALPHABET)
            else:  # decrypt
                idx = (idx - key) % len(ALPHABET)
            result += ALPHABET[idx]
        else:
            result += ch
    return result

while True:
    print("\n==== Caesar Cipher ====")
    print("1. Encrypt a Message")
    print("2. Decrypt a Message")
    print("3. Quit")

    choice = input("Choose an option (1/2/3): ").strip()

    if choice == "3":
        print("Goodbye!")
        break
    elif choice not in ["1", "2"]:
        print("Invalid choice. Try again.")
        continue

    try:
        key = int(input(f"Enter key (0 to {len(ALPHABET)-1}): "))
        if not (0 <= key < len(ALPHABET)):
            print("Invalid key range.")
            continue
```

```
except ValueError:
    print("Key must be a number.")
    continue

message = input("Enter your message: ")
mode = "encrypt" if choice == "1" else "decrypt"
result = caesar_cipher(message, key, mode)

print("Result:", result)
```

5. Example Run

Encrypt:
==== Caesar Cipher ====
1. Encrypt a Message
2. Decrypt a Message
3. Quit
Choose an option (1/2/3): 1
Enter key (0 to 39): 3
Enter your message: HELLO
Result: KHOOR

Decrypt:
==== Caesar Cipher ====
1. Encrypt a Message
2. Decrypt a Message
3. Quit
Choose an option (1/2/3): 2
Enter key (0 to 39): 3
Enter your message: KHOOR
Result: HELLO

6. Conclusion

- The Caesar Cipher is a simple yet effective method to demonstrate encryption and decryption.
- This program successfully allows the user to input a message and key, and then performs encryption or decryption based on user choice.
- It meets the requirements of Task – 01 by implementing Caesar Cipher in Python.